Crossing the Steel River

BUILD · Humanities · 40 minutes

I can match a river-crossing problem to two different steel bridge solutions.

Name: ___________________________ Date: ______________

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

We do · make a choice and give a reason

Commit before reveal: Which Transporter feature kept a road deck out of the channel?

- A suspended moving gondola
- Blue paint
- A fixed low roadway

My choice and the evidence I used:

Resources and independent record

BUILD Humanities · W13 · Crossing the Steel River

Enquiry: How did engineers help people cross a busy working River Tees?

Learning objective: I can match a river-crossing problem to two different steel bridge solutions.

Inherited: Week 12 growth meant more people, work and movement near the river.

CONSTRUCTED TEACHING SOURCE — created for classroom practice. It is not an archival original.

Independent Humanities mission

A dated bridge comparison using need, constraint, mechanism and one evidence limitation.

Evidence target: A dated bridge comparison using need, constraint, mechanism and one evidence limitation.

Supported

Match six bridge cards.

Evidence: Matched board plus one because-link.

Scaffold: The ___ Bridge used ___.

Standard

Complete a need, constraint and solution matrix for both bridges.

Evidence: Comparison matrix.

Scaffold: People needed ___, but ships needed ___, so ___.

Stretch

Judge how each solved its period’s problem and add a source limit.

Evidence: Comparative judgement.

Scaffold: Both ___. Unlike ___. The evidence cannot tell ___.

Source/material record

Teacher-prepared constructed lab cards plus the locally approved source cards named for this lesson.

Historian/geographer response & Influence

Pupil Humanities decision:

Evidence or representation used:

Reasoning, limitation or uncertainty:

Audience genuinely received: describe through the agreed school workflow; no signature is required here.

Influence: The named class teacher opens Week 14 through the chosen lens while still showing both lenses in the lesson.

Decision maker: Class teacher — replace with the adult’s name before delivery.

Report back by: Start of Week 14 — add the calendar date before delivery.

Learner source cards

ADAPTED SOURCE CARD

HUM-B-W13-C01 • Transporter Bridge: stable facts

Provenance source: HUM-B-W13-S01

Middlesbrough Council states that the Tees Transporter Bridge between Middlesbrough and Port Clarence opened in 1911. Its gondola was intended to carry vehicles and pedestrians across the River Tees.

Can establish Establish opening year, location and intended gondola function.

Cannot establish Establish current operating status or every reason for choosing the design.

Accessible route Three labels: 1911; high frame; suspended gondola carries people and vehicles across.

HUM-B-W13-C02 • Newport Bridge: stable facts

Provenance source: HUM-B-W13-S03

Historic England records Newport Bridge as built in 1931–34 with a vertically lifting centre road span. When raised, it provided 36 metres of clearance for river traffic at high water.

Can establish Establish date, vertical-lift mechanism and recorded clearance.

Cannot establish Establish current lifting capability or every reason for selecting this design.

Accessible route Two-frame description: road span down for crossing; centre span raised for 36 m high-water clearance.

HUM-B-W13-C03 • Constructed comparison brief

Provenance source: HUM-B-W13-S02

Classroom brief: people and vehicles need to cross, while river traffic may need clearance. Compare how each moving mechanism could respond. This brief is an evidence-informed inference, not a quotation from the engineers.

Can establish Support structured comparison of the documented mechanisms.

Cannot establish Prove the complete historical decision process behind either bridge.

Accessible route Need → possible constraint → mechanism: cross the river → allow clearance → gondola across or centre span up.

Source provenance

HUM-B-W13-S01 · Tees Transporter Bridge Maker: Middlesbrough Council · Date: Page updated 5 August 2025 Origin: Middlesbrough Council transport and heritage information Rights/use: All rights reserved. Stable facts paraphrased; no council video or image embedded. Access: Original labelled gondola diagram with people, vehicles and river; equivalent text.

HUM-B-W13-S02 · Transporter Bridge — entry 1139267 Maker: Historic England · Date: Structure dated 1911 Origin: National Heritage List for England; Historic England Rights/use: Official list-entry text OGL v3.0 with Historic England attribution. Maps and user/archive images excluded. Access: Simple side elevation with labelled suspended carriage and three-sentence description.

HUM-B-W13-S03 · Newport Bridge — entry 1139837 Maker: Historic England · Date: Built 1931–34 Origin: National Heritage List for England; Historic England Rights/use: Official list-entry text OGL v3.0 with Historic England attribution. Map and archive/contributed images excluded. Access: Original two-frame down/raised diagram with movement arrows and text alternative.

Print surface requires physical classroom checking before live use.

Paper route • discuss the lesson choices

Use these choices with the matching teaching stage. Point to or number a choice, explain it, then revisit it after feedback. The original HTML keeps the live controls.

- Transporter Bridge
- Newport Bridge
- opened 1911
- opened 1934
- a gondola hangs from a high frame and glides across
- the whole road deck lifts up between two towers
- ships pass under the frame while the gondola waits
- traffic stops while the deck is raised for a ship
- What can the diagram NOT show?
- Gondola at Middlesbrough side
- Gondola travelling beneath high frame
- Gondola at Port Clarence side
- Centre road span down for road crossing
- Centre road span rising
- Centre road span raised with 36 m high-water clearance
- TRANSPORTER 1911
- TRANSPORTER suspended gondola
- TRANSPORTER Middlesbrough–Port Clarence
- NEWPORT built 1931–34
- NEWPORT lifting central road span
- NEWPORT 36 m clearance when raised
- BOTH steel river crossing
- BOTH moving clearance solution

Choice / card	My reason or evidence	My revision after feedback

Exit • one next step

After the adult has listened, what will you keep, improve or check next?